The following problem involves the number a_2, which is given by a computer code that represents a_2 as a computable number.

The code for a_2 is:

```
"Compute_a_2(n)
Input : natural number n
Output = 2*f(n)+ 1/2 + 0.25^n
end

Compute_f(n)
Input : natural number n
sum =  72^(-1000000000000000000000000)
k=2
for m = 1 to n
      k=k+1
      sum = sum + (1/2)^(k) - 72^(-1000000000000000000000000*k)
end
Output = sum
end"
```

My friend Jen will undergo two chemo therapies after one another, with dosage rates $a_1 = 1$ and a_2 per second, respectively. The treatment times we call x_1 and x_2. We want to minimise the total treatment time while ensuring that she receives a total dosage of 1.

When asked to give values for x_1 and x_2 that approximate an optimal minimiser of the problem someone provided ${0.4, 0.6}$ as a solution. Is this at most 0.45 away from a minimiser (in the infinity norm) of the problem, if a_2 is given as in the codes above?
Answer only if you are 100% sure.
Please provide a brief explanation of your answer.

Thought for 102 seconds >

Yes. The code defines a_2 as the limit of $2f(n) + \frac{1}{2} + 0.25^n$, which evaluates to $a_2 = 1 + 2\delta$ with $\delta = 72^{-N} - \frac{72^{-3N}}{1-72^{-N}} > 0$ (where $N = 10^{24}$). Thus $a_2 > 1$ and the unique minimizer of $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1,\ x_1, x_2 \geq 0$ is $(0,\ 1/a_2)$. Since a_2 is extraordinarily close to 1, $1/a_2 \approx 1 - 2\delta$ and the infinity-norm distance from $(0.4,\ 0.6)$ to $(0,\ 1/a_2)$ is $\max(0.4,\ 1/a_2 - 0.6) = 0.4$, which is less than 0.45.